

Chess Agent Competition Track

Project Information Sheet

Course	ECE-GY 6263 Game Theory, Fall 2026
Project category	Option 3 Original Research or Computational Project
Format	Individual project

The Chess Agent Competition is a structured computational track under Option 3. Each student develops one resource-bounded chess agent, evaluates it experimentally, and explains its strategic design in the course proposal, presentations, and final report. A common benchmark and a student-versus-student tournament provide shared evidence, but chess performance is only one part of the project grade.

1. Project overview and learning objectives

Each submitted agent is a strategy: given a chess position, limited information, and a clock, it must choose a legal move. The project connects implementation choices with bounded rationality, empirical payoffs, adaptation, dominance, exploitability, and nontransitive interaction. It fits Option 3 because students build a computational artifact and use it to investigate a game-theoretic question.

By completing the project, students should be able to:

- implement a valid decision policy under time, memory, dependency, and information limits;
- compare search, evaluation, randomization, time-management, or opponent-modeling methods;
- design reproducible experiments and at least one meaningful controlled comparison or ablation;
- interpret head-to-head results as an empirical strategic population rather than only a win count;
- analyze limitations, failure cases, exploitability, and performance-reliability tradeoffs;
- communicate the method and evidence in a technical report and presentation.

2. Student work and deliverables

Each student submits one Python file containing exactly one no-argument subclass of the supplied `ChessAgent` interface. On each turn, the agent receives a structured observation containing the current position, move history, clock information, color, a reproducible seed, and a matchup-local anonymous opponent identifier. It returns one legal move in UCI notation.

Students may use methods such as minimax or negamax, alpha-beta pruning, positional evaluation, move ordering, iterative deepening, quiescence search, seeded randomized policies, time management, and bounded opponent modeling. External chess engines, network services, unauthorized packages, and additional model or data files are prohibited.

Deliverable	Date	Requirement
Project proposal	October 17, 2026, 5:00 PM ET	Approximately 1-2 pages describing the question, preliminary design, experiments, milestones, and references
Midterm presentation	October 30, 2026	Architecture, strategic hypothesis, preliminary evidence, risks, and remaining plan
Final agent submission	November 23, 2026, 5:00 PM ET	One validated NetID-named Python file submitted through Gradescope
Final presentation	December 4, 2026	Final method, central findings, competition results available at that time, and interpretation
Final report	December 19, 2026, 5:00 PM ET	At least 10 pages in IEEE double-column style

The final-agent deadline freezes the version used for the official benchmark, Student League, and playoffs. Any schedule change will be announced through the usual course channels.

The final report must state the problem and hypotheses, explain the agent and implementation, present controlled experiments, analyze local and official results, discuss strategic interactions and threats to validity, and propose future work. Students must acknowledge all authorized borrowed ideas and code.

3. Development and submission workflow

Students receive a local starter ZIP containing the versioned API, a template, public Random and Greedy agents, a shallow minimax example, a validator, a local match runner, and PGN replay reports. The launchers create a project-local environment from the supplied Python and dependency lock files. A browser interface runs only on the student's computer through 127.0.0.1; it is not a course upload website.

```

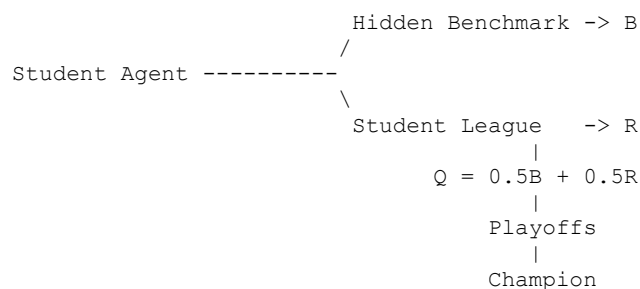
Download and extract the starter ZIP
|
create YOUR_NETID.py
|
validate and run local matches
|
submit only YOUR_NETID.py to Gradescope
|
TA freezes the file and its SHA-256

```

Local tests help students find interface errors and compare versions against public practice agents. They do not reveal the official hidden benchmark score or class rank. The [Student Guide](#) contains setup, API, validation, and submission instructions.

4. Official competition

Every technically valid submission enters the same qualification process.



The Hidden Benchmark Score B comes from 32 games against four instructor-side variants: two Standard and two Advanced. Games use paired starting positions with colors reversed. Public practice agents do not count toward B .

The Student League Score R comes from a complete class round robin. Each pair of students plays four games using two frozen legal starting positions. Each position is played twice with colors reversed, and the order gives both agents one early and one late game as White and as Black. This preserves color balance while making the four games more informative for deterministic and matchup-adaptive agents.

For two or more valid submissions, $Q = 0.5B + 0.5R$. If there is only one valid submission, $Q = B$ and no class champion is declared. Q determines qualification and seeding.

Valid submissions N	Playoff field K
$N \geq 32$	16
$16 \leq N < 32$	8
$8 \leq N < 16$	4
$2 \leq N < 8$	2
$N < 2$	No championship bracket

The bracket is single elimination with seeded pairings and color-reversed starting-position pairs. Detailed clocks, fault handling, matchup memory, qualification tiebreaks, playoff series, hidden and public information, and placement formulas are specified in the [Public Competition Rules](#).

5. Grading

Term Project component	Points
Tournament Performance	10
Correctness reliability and reproducibility	15
Agent methodology and implementation	25
Experimental and game-theoretic analysis	30
Final report quality limitations and future work	20
Total	100

Tournament Performance is one 10-point component. The overall project grade also evaluates methodology, implementation, reliability, analysis, and the final report, so the tournament champion does not automatically receive the highest overall project grade.

Under the Fall 2026 syllabus, the Term Project contributes 60 percent of the course grade and the Presentation contributes 40 percent. The table above allocates the Chess Track Term Project grade and does not change those syllabus-level weights.